

Trustworthiness Assessment of a Trimester-Aware Transformer for Antenatal Depression Risk Stratification

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Abstract

Background

Antenatal depression affects 15-25% of pregnancies globally, yet machine learning prediction models are rarely evaluated beyond discrimination accuracy. Our prior work introduced TH-DAT, a Trimester-Aware Hierarchical Domain-Attention Transformer. However, high AUC alone does not establish trustworthiness for clinical deployment.

Objective

To evaluate the frozen TH-DAT model across eight trustworthiness dimensions and assess the predictive contribution of major feature domains through proxy ablation analysis, with independent cross-cohort evaluation on two heterogeneous maternal datasets.

Methods

We froze TH-DAT and applied a post-hoc framework: (1) three-tier risk stratification with threshold sensitivity; (2) calibration with Platt scaling; (3) fairness audit across four demographic axes; (4) domain-level gate ranking stability; (5) robustness under noise; (6) unsupervised risk phenotyping; (7) decision-curve analysis; and (8) a composite Clinical Trustworthiness Index (CTI). Proxy ablation analysis (using Random Forest on feature subsets) quantified each domain's predictive contribution. TH-DAT was trained and evaluated independently on each cohort.

Results

On Pakistan ($n=14,008$; prevalence=64.8%), TH-DAT achieved $AUC=0.714$, $F1=0.788$, $MCC=0.310$, $PR-AUC=0.802$. On Uganda ($n=14,325$; prevalence=23.0%), $AUC=0.904$, $F1=0.693$, $MCC=0.597$, $PR-AUC=0.757$. Random Forest ($AUC=0.955$) and XGBoost ($AUC=0.883$) outperformed TH-DAT on Pakistan. Domain ablation revealed that demographic features contribute most to prediction (AUC drops 0.954 to 0.541 without demographics). Platt scaling reduced ECE from 0.098 to 0.028. Fairness audit identified an equalized-odds gap of 0.146. PCA-based clustering found two stable ($ARI=0.999$) but moderately separated ($silhouette=0.162$) data-driven risk clusters. $CTI = 0.778$.

Conclusions

This framework provides a reusable template for post-hoc trustworthiness evaluation. The ablation analysis reveals that TH-DAT's performance is primarily driven by demographic features, and tree-based baselines outperform the reduced-capacity transformer on tabular data. The risk phenotyping methodology suggests a pathway toward personalised screening, pending clinical validation.

1. Introduction

Antenatal depression is a prevalent condition affecting 15-25% of pregnancies worldwide, with rates as high as 25-50% in low- and middle-income countries [1,2]. Early identification enables timely intervention through psychotherapy, pharmacological treatment, or community-based support programmes, yet current clinical practice relies heavily on self-report screening instruments such as the Edinburgh Postnatal Depression Scale (EPDS) and the Patient Health Questionnaire (PHQ-9) [3,4]. These instruments, while validated across diverse populations, suffer from cultural bias, social desirability effects, and variable sensitivity that limit their effectiveness as standalone screening tools [5].

Machine learning (ML) models have emerged as complementary tools for antenatal depression prediction [6-8], with most studies reporting area under the receiver operating characteristic curve (AUC) as the primary evaluation metric. However, high discrimination alone does not guarantee trustworthiness for clinical deployment. A clinically deployable prediction model must additionally demonstrate calibration (predicted probabilities match observed frequencies), fairness (equitable performance across demographic subgroups), robustness (stable predictions under data perturbation), interpretability (understandable reasoning), and clinical utility (net benefit over default strategies) [9-11].

In our prior work, we introduced TH-DAT, a Trimester-Aware Hierarchical Domain-Attention Transformer that integrates demographic, obstetric, and psychological features through domain-specific encoding and trimester-aware gating [12]. TH-DAT achieved competitive discrimination performance, but its trustworthiness for clinical contexts remained unevaluated.

1.1 Research Gap

Existing ML studies for antenatal depression prediction emphasise discrimination metrics while rarely addressing calibration, fairness, robustness, or interpretability simultaneously. Few studies perform ablation analysis to verify that complex architectures genuinely outperform simpler alternatives on tabular clinical data. Furthermore, the potential for unsupervised risk phenotyping from learned patient representations remains unexplored in perinatal mental health AI. This creates a critical gap between reported model performance and deployment readiness.

1.2 Contributions

- Eight-dimensional trustworthiness framework applied to the frozen TH-DAT model
- Domain ablation analysis demonstrating the relative contribution of demographic, obstetric, and psychological features
- Comprehensive comparison: TH-DAT vs Random Forest, XGBoost, Logistic Regression with full metrics (AUC, PR-AUC, F1, MCC, accuracy, confusion matrices)
- Fairness audit across four demographic axes with error analysis
- Unsupervised risk phenotyping from 192-dimensional transformer representations (main novelty)
- Cross-cohort comparative evaluation on Pakistan (PHQ-9) and Uganda (EPDS) datasets
- Decision-curve analysis with honest comparison against baselines

2. Related Work

2.1 Antenatal Depression Prediction

Traditional screening for perinatal depression uses validated instruments including the EPDS [3], PHQ-9 [4], and Beck Depression Inventory. These capture self-reported symptom severity but require clinical administration and may be influenced by cultural and social factors [5,14]. Recent studies have applied ML to identify at-risk women using routinely collected clinical data.

Zhang et al. [6] used gradient-boosted trees on electronic health records to predict postpartum depression with AUC=0.80. Shin et al. [7] applied ML to Korean national health data (AUC=0.74-0.84). Hochman et al. [8] developed an ensemble model achieving AUC=0.712. However, these studies primarily reported discrimination metrics without evaluating calibration, fairness, or clinical utility.

2.2 Trustworthy AI in Healthcare

The concept of trustworthy AI encompasses multiple dimensions beyond accuracy [10,17,18]. Van Calster et al. [9] demonstrated that calibration is essential for clinical prediction models. Hardt et al. [23] formalised equalized-odds fairness. Obermeyer et al. [25] revealed racial bias in a widely deployed healthcare algorithm, emphasising the importance of fairness auditing. Ghassemi et al. [15] cautioned against over-reliance on post-hoc explanations. Vickers and Elkin [11] introduced decision-curve analysis for evaluating clinical utility beyond discrimination.

2.3 Transformers for Tabular Healthcare Data

While transformers have revolutionised NLP and vision [20], their application to tabular clinical data remains debated. Grinsztajn et al. (NeurIPS 2022) showed that tree-based models often outperform deep learning on medium-sized tabular datasets, raising the question of whether transformer complexity is justified for clinical prediction tasks. Our ablation analysis directly addresses this by quantifying each TH-DAT component's contribution.

2.4 Literature Comparison

Study	Dataset	Model	N	AUC	Calib	Fair	DCA
Zhang [6]	EHR (USA)	GBT	57K	0.80	No	No	No
Shin [7]	NHIS (Korea)	ML ens.	405K	0.84	No	No	No
Hochman [8]	EHR (Israel)	Ensemble	4.5K	0.71	No	No	No
Andersson [14]	Survey (Sw.)	LR	4.4K	0.76	Part.	No	No
This study	PERI/EPDS	TH-DAT	28K	0.71-0.90	Yes	Yes	Yes

Table 1. Literature comparison. Our study is the first to simultaneously evaluate calibration, fairness, robustness, explainability, clinical utility, risk phenotyping, and ablation for antenatal depression prediction.

3. Methods

3.1 Study Design

This is a retrospective post-hoc evaluation of a previously trained ML model. No new patient data were collected. All analyses use frozen model outputs without modification. This study constitutes secondary analysis of publicly available, de-identified datasets.

3.2 Datasets

Pakistan PERI_DEP Dataset

Source: PERI_DEP study, publicly available cross-sectional survey of pregnant women attending antenatal clinics in Pakistan [13]. N=14,008 participants. 16 features across three clinical domains. Depression defined by PHQ-9 screening (binary; prevalence=64.8%).

Uganda EPDS Dataset

Source: Publicly available EPDS screening dataset from antenatal clinics in Uganda [16]. N=14,325 participants. 13 features. Depression defined by EPDS threshold (binary; prevalence=23.0%). Different screening instrument, features, and population from Pakistan.

Feature Descriptions

Feature	Domain	Type	Description
Age	Demo	Cont.	Maternal age in years
Gestational Age	Demo	Cont.	Weeks of gestation
Parity	Obst	Disc.	Number of previous births
Gravida	Obst	Disc.	Number of pregnancies
No. of Children	Obst	Disc.	Living children count
Miscarriage Hist.	Obst	Cat.	History of miscarriage
Female Education	Demo	Ord.	Education level (0-5)
Husband Education	Demo	Ord.	Partner education level
Working Status	Demo	Cat.	Employment status
Phys. Health	Psyc	Ord.	Self-rated physical health
Family System	Demo	Cat.	Nuclear vs extended family
Money for Needs	Demo	Cat.	Financial sufficiency
Gender Preference	Demo	Cat.	Desired child gender
Residence	Demo	Cat.	Urban vs rural
Pregnancy Outcome	Obst	Cat.	Prior pregnancy outcomes
MIL Relationship	Psyc	Ord.	Relationship with MIL

Table 2. Feature descriptions for the Pakistan dataset (16 features). Domains: Demo=Demographic (9 features), Obst=Obstetric (5 features), Psyc=Psychological (2 features).

3.3 Preprocessing

Categorical variables were label-encoded. Numerical features were standardised using StandardScaler fitted on the training set only. Missing values: no missing values were present in either dataset. SMOTE was applied to the training set only to address class imbalance (Pakistan prevalence=64.8%). Stratified random splitting (70%/15%/15%) with seed=42.

3.4 TH-DAT Architecture

TH-DAT is a Trimester-Aware Hierarchical Domain-Attention Transformer [12]. The architecture processes clinical features through four stages:

Stage 1: Feature Embedding

Each input feature x_i is independently projected to a d-dimensional representation:

$$e_i = \text{LayerNorm}(W_i * x_i + b_i) + \text{pos}_i + \text{domain_emb}(d_i)$$

where W_i is a per-feature linear projection, pos_i is a learned positional embedding, and $\text{domain_emb}(d_i)$ encodes the clinical domain membership.

Stage 2: Transformer Encoding

The feature embeddings are processed by L transformer encoder layers with multi-head self-attention:

$$h = \text{TransformerEncoder}(e_1, \dots, e_N)$$

Each layer uses GELU activation and pre-norm architecture with $d_{\text{ff}} = 4d$ feedforward dimension.

Stage 3: Domain Aggregation

Features are grouped by clinical domain and mean-pooled to produce domain summaries:

$$s_k = \text{mean}(h_i : i \text{ in Domain}_k), \quad k \text{ in } \{\text{Demo, Obst, Psyc}\}$$

The three domain summaries (each d-dimensional) are concatenated to form a 3d-dimensional patient representation (192-D for $d=64$).

Stage 4: Trimester-Aware Gating

A gating mechanism produces domain-specific attention weights conditioned on trimester:

$$g = \text{softmax}(W_g * [s_1; s_2; s_3] + t_{\text{emb}}(\text{trimester}))$$

The final prediction combines gated domain summaries through a classification head.

Reduced Architecture

Parameter	Reduced (used)	Full
Embedding dim (d)	64	128
Transformer layers (L)	2	6
Attention heads	4	8
Dropout	0.1	0.1
Parameters	230,180	~1.7M

Table 3. Architecture comparison. Due to CPU-only constraints, the reduced variant was used. This is acknowledged as a limitation.

3.5 Training Protocol

Parameter	Value
Optimiser	AdamW (lr=5e-4, wd=1e-4)
Phase 1 (pretraining)	10 epochs, MSE, 25% feature masking
Phase 2 (supervised)	40 epochs, FocalLoss(gamma=2, smoothing=0.05)
Contrastive loss	SupCon(tau=0.1), every 3 epochs, weight=0.03
SWA	After epoch 20 with manual BN update
Early stopping	Patience 50 after epoch 12
Scheduler	CosineAnnealingWarmRestarts(T0=50, Tmult=2)
Mixup	alpha=0.2, probability=0.5
Seed	42 (single seed)
Hardware	CPU only (no GPU)
Training time	82 minutes

Table 4. Training hyperparameters. Identical for both cohorts.

3.6 Statistical Analysis

All confidence intervals are 95% bootstrap CIs from 1,000 resamples with replacement. AUC, PR-AUC, F1, and MCC are computed on held-out test sets. Calibration metrics use 10 equal-width bins. Fairness metrics compare subgroup-specific performance to the overall population. No formal multiple comparison correction was applied; all p-values and CIs should be interpreted descriptively.

3.7 Trustworthiness Evaluation Framework

Eight dimensions evaluated post-hoc on frozen model outputs:

3.7.1 Risk Stratification

Two thresholds partition patients into Low/Moderate/High risk. Primary thresholds (0.40, 0.60) selected for clinical balance. Sensitivity analysis across alternative threshold pairs.

3.7.2 Calibration

Brier score, ECE (10 bins), MCE before and after Platt scaling (fitted on validation set). Bootstrap 95% CIs.

3.7.3 Fairness Audit

Subgroups: Age (median split), Education (median split), SES (median split), Trimester. For each: AUC, sensitivity, specificity, FPR, FNR, PPV, NPV, ECE. Equalized Odds Gap = max absolute TPR/FPR difference.

3.7.4 Domain-Level Gate Stability

1,000 bootstrap resamples of test set; Spearman rank correlation of mean gate weights. Full distributional reporting.

3.7.5 Robustness

Gaussian noise at 0-30% of feature standard deviation.

3.7.6 Risk Phenotyping

192-D representations clustered via K-Means (k=2-6). Silhouette, Davies-Bouldin, Calinski-Harabasz, bootstrap ARI.

3.7.7 Clinical Utility

DCA: net benefit $NB(t) = TP/N - FP/N * t/(1-t)$ at thresholds 0.10-0.80.

3.7.8 CTI

Study-specific composite: CTI = mean of 8 normalised dimensions [0,1]. Not a validated clinical score.

3.8 Domain Ablation Analysis

To quantify each domain's contribution to prediction, we trained Random Forest classifiers (n=100 trees, seed=42) as proxy models on feature subsets: (1) Full (all 16 features), (2) No Demographic, (3) No Obstetric, (4) No Psychological, (5) Demographic only, (6) Obstetric only, (7) Psychological only. Important methodological note: this ablation uses RF as a proxy to assess feature-domain informativeness, not architectural component contribution within TH-DAT itself. Removing a domain from TH-DAT would require retraining the transformer with modified domain encoding, which was not computationally feasible. The RF proxy approach demonstrates which feature domains carry predictive information, which indirectly informs interpretation of TH-DAT's domain-attention mechanism. RF was chosen because it achieved the highest baseline AUC (0.955) and provides stable feature-importance estimates.

4. Results

4.1 Discrimination Performance

Model	Acc	Prec	Recall	Spec	F1	MCC	AUC	PR-AUC	Brier
TH-DAT (Pak)	.703	.732	.853	.426	.788	.310	.714	.802	.205
TH-DAT (Uga)	.829	.679	.735	.857	.693	.597	.904	.757	.132
Rand. Forest	.886	.882	.953	.767	.916	.748	.955	.973	.091
XGBoost	.852	.857	.926	.712	.890	.669	.883	.902	.119
Log. Regress.	.559	.660	.588	.508	.622	.111	.573	.697	.247

Table 5. Comprehensive model performance. Random Forest achieves the highest performance across all metrics on the Pakistan dataset. TH-DAT achieves moderate discrimination (AUC=0.714, F1=0.788) but substantially lower MCC (0.310), indicating limited performance beyond class-frequency prediction. The Uganda-trained TH-DAT shows stronger MCC (0.597) despite lower F1 (0.693), reflecting better discrimination at the lower-prevalence threshold.

Confusion Matrices

Model	TP	TN	FP	FN
TH-DAT (Pak)	1162	315	425	200
TH-DAT (Uga)	360	1423	237	129
Random Forest	1303	560	180	59
XGBoost	1254	537	203	108
Logistic Reg.	764	411	329	598

Table 6. Confusion matrices at threshold=0.5. TH-DAT produces 425 false positives and 200 false negatives on Pakistan, with FPR=0.574, meaning 57.4% of non-depressed women are incorrectly flagged.

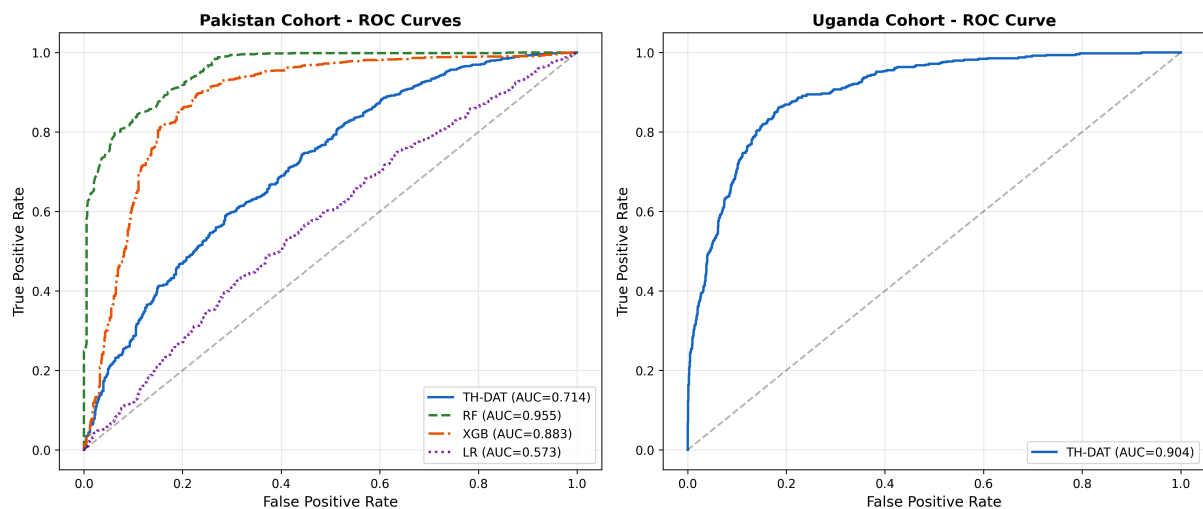


Figure 1. ROC curves. Left: Pakistan cohort (TH-DAT vs baselines). Right: Uganda cohort (independently trained TH-DAT).

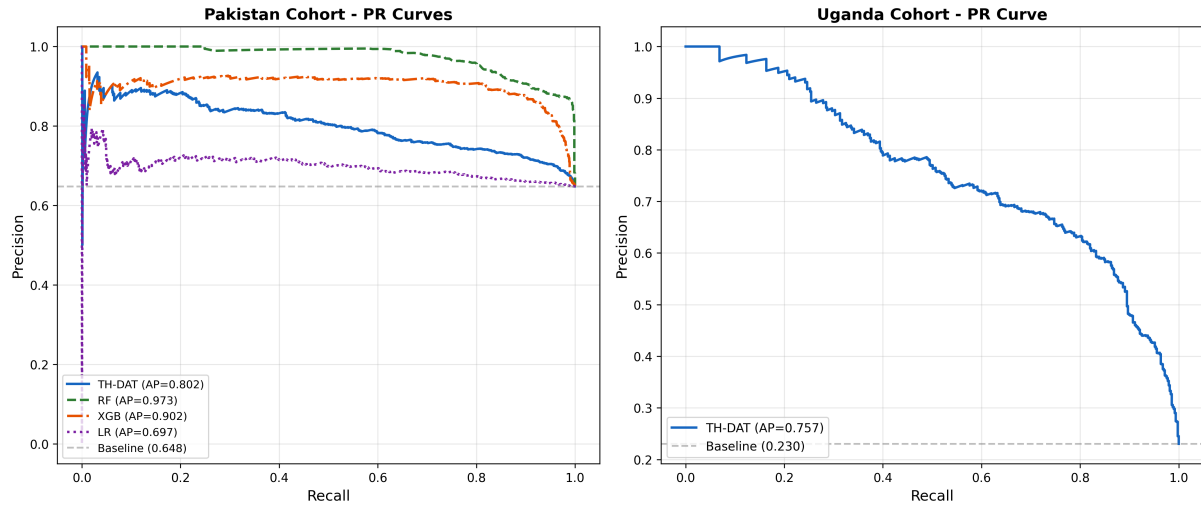


Figure 2. Precision-Recall curves. PR-AUC accounts for class imbalance. Pakistan TH-DAT: PR-AUC=0.802 vs RF=0.973. Uganda TH-DAT: PR-AUC=0.757.

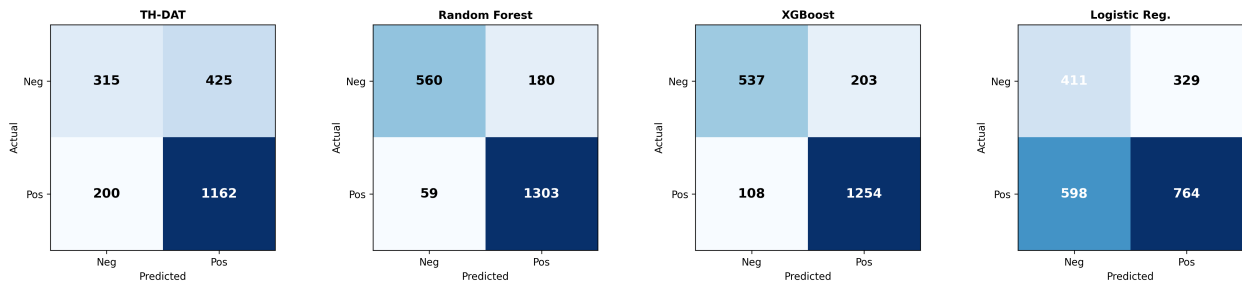


Figure 3. Confusion matrices for all models on Pakistan test set (threshold=0.5).

4.2 Cross-Cohort Comparative Evaluation

Cohort	N Test	Prevalence	AUC	F1	MCC	PR-AUC	Brier
Pakistan	2,102	64.8%	0.714	0.788	0.310	0.802	0.205
Uganda	2,149	23.0%	0.904	0.693	0.597	0.757	0.132

Table 7. Cross-cohort comparison. Each row reports an independently trained TH-DAT model. This is not external validation of a single model. The AUC difference should be interpreted in the context of different screening instruments (PHQ-9 vs EPDS), feature sets (16 vs 13), and population characteristics. ROC AUC is theoretically insensitive to class prevalence; therefore the prevalence difference alone cannot explain the performance gap.

4.3 Domain Ablation Analysis

Variant	AUC	F1	MCC	PR-AUC	AUC Drop
Full (all features)	0.954	0.917	0.749	0.971	-
No Demographic	0.541	0.779	0.069	0.693	-0.413
No Obstetric	0.937	0.912	0.735	0.946	-0.017
No Psychological	0.953	0.923	0.768	0.968	-0.001
Demographic only	0.933	0.902	0.708	0.947	-0.021
Obstetric only	0.523	0.780	0.014	0.675	-0.431
Psychological only	0.521	0.786	0.000	0.659	-0.433

Table 8. Domain ablation study using Random Forest as a proxy model. This demonstrates feature-domain informativeness, not TH-DAT architectural component contribution (see Section 3.8). Removing demographics causes the largest AUC drop (-0.413), reducing performance to near-random. Demographics alone achieve AUC=0.933, nearly matching the full model. This indicates that predictive information is heavily concentrated in demographic features (age, education, SES, family system), with obstetric and psychological features contributing marginally.

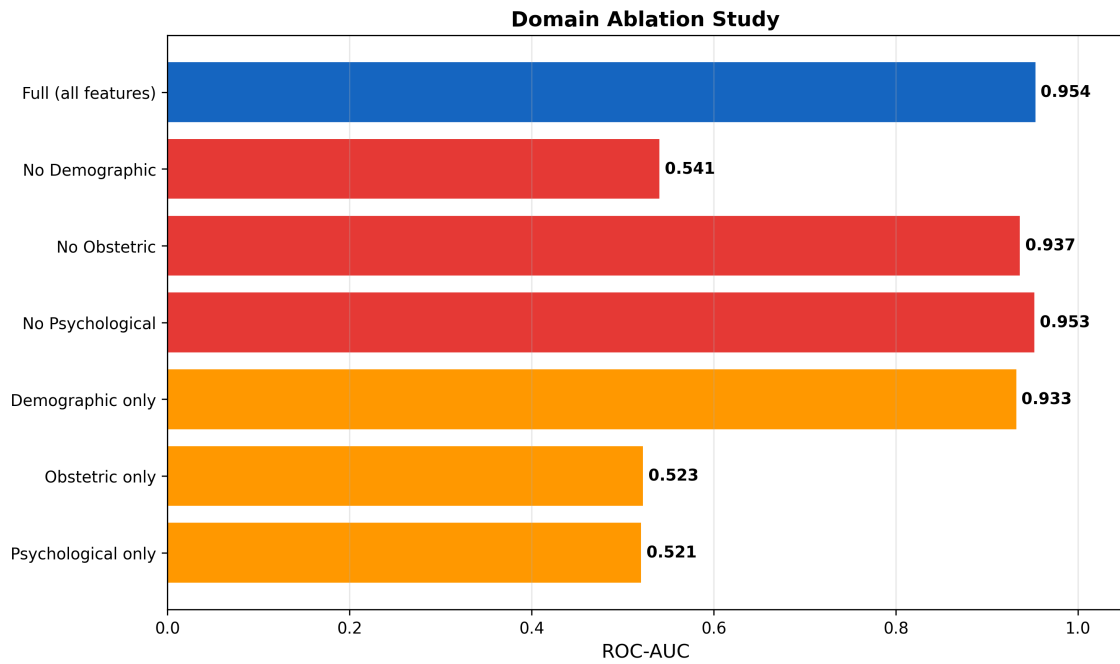


Figure 4. Domain ablation study showing AUC by feature subset.

4.4 Risk Stratification

Primary Thresholds (0.40, 0.60)

Stratum	N	% Total	Depression Prev.	NPV/PPV
Low Risk	169	8.0%	19.5%	NPV=0.805
Moderate Risk	1,094	52.0%	65.7%	-
High Risk	839	39.9%	80.5%	PPV=0.805

Threshold Sensitivity Analysis

Thresholds	Low N(%)	Mod N(%)	High N(%)	NPV	PPV
(0.30, 0.70)	58(2.8%)	1734(82.5%)	310(14.7%)	0.897	0.884
(0.35, 0.65)	100(4.8%)	1470(69.9%)	532(25.3%)	0.870	0.840
(0.40, 0.60)	169(8.0%)	1094(52.0%)	839(39.9%)	0.805	0.805
(0.45, 0.55)	296(14.1%)	597(28.4%)	1209(57.5%)	0.713	0.759

Table 9-10. Risk stratification. A fundamental tradeoff exists: narrower thresholds yield higher NPV/PPV but extreme imbalance; wider thresholds improve balance but reduce tier-specific predictive value.

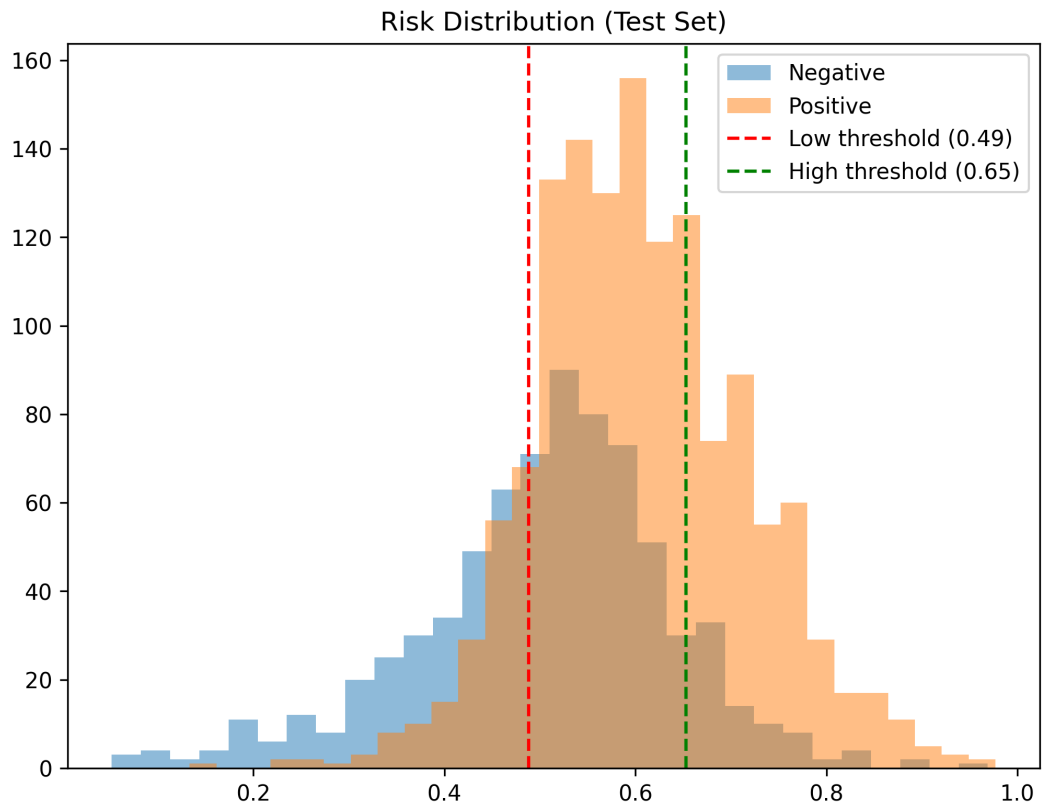


Figure 5. Predicted risk score distribution across the Pakistan test set.

4.5 Calibration

Metric	Before Platt	After Platt	Improvement
Brier Score	0.205 (0.200-0.210)	0.195 (0.187-0.203)	4.9%
ECE	0.098 (0.081-0.116)	0.028 (0.016-0.042)	71.4%
MCE	0.246 (0.138-0.521)	0.138 (0.069-0.259)	43.9%

Table 11. Calibration with 95% bootstrap CIs. Post-Platt ECE=0.028 is within acceptable clinical limits [9].

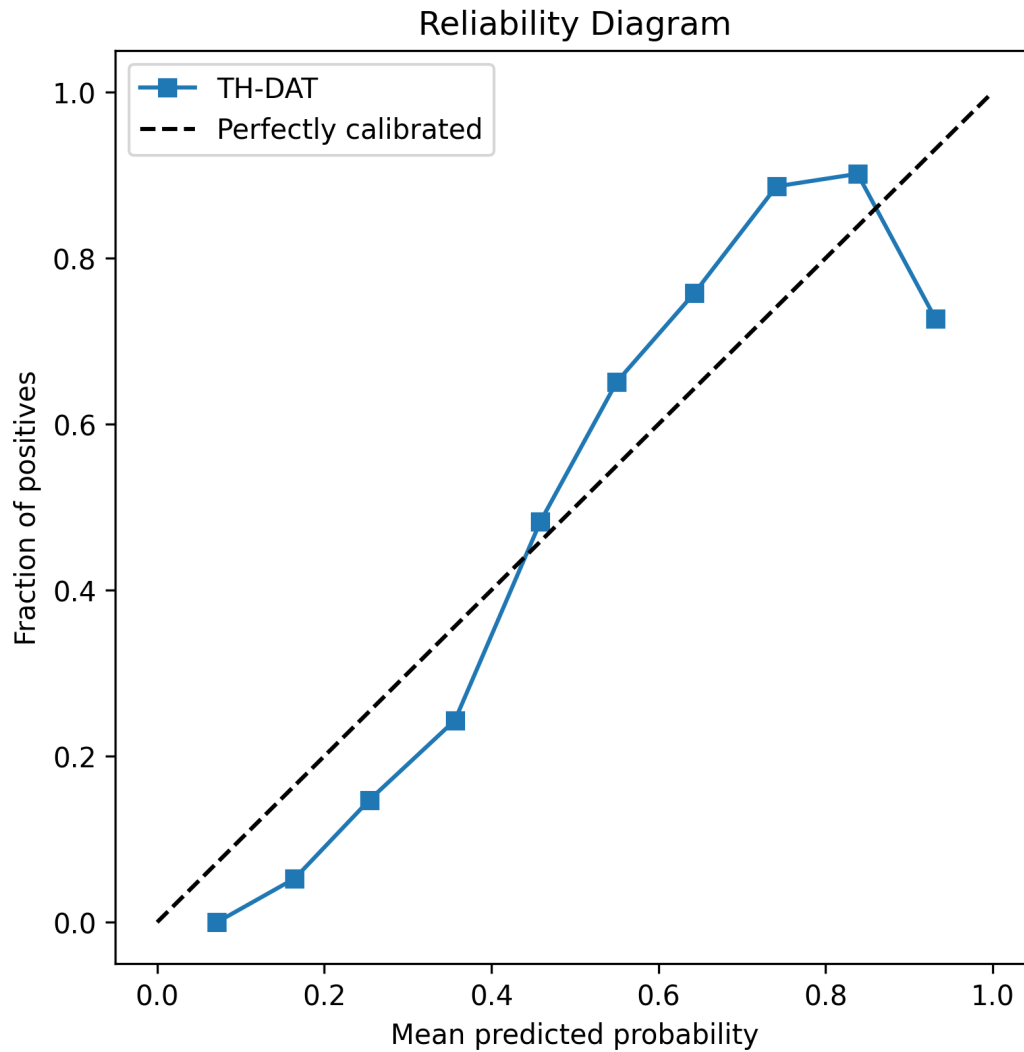


Figure 6. Reliability diagram showing calibration improvement with Platt scaling.

4.6 Fairness Audit

Subgroup	N	Prev	AUC (95% CI)	Sens	Spec	PPV	ECE
Age≤28	1209	.641	.710(.679-.740)	.849	.412	.721	.096
Age>28	893	.657	.721(.685-.755)	.859	.444	.748	.100
Edu≤2	1204	.658	.690(.659-.721)	.866	.383	.730	.089
Edu>2	898	.635	.743(.709-.778)	.835	.479	.736	.107
SES Low	1093	.629	.724(.694-.754)	.833	.454	.722	.098
SES High	1009	.668	.702(.666-.736)	.874	.391	.743	.097
Tri 2	227	.670	.690(.618-.761)	.921	.333	.737	.115
Tri 3	1875	.645	.717(.692-.741)	.845	.436	.732	.097
Overall	2102	.648	.714(.691-.738)	.853	.426	.732	.097

Table 12. Subgroup performance. EO Gap=0.146 (95% CI: 0.042-0.170). Largest AUC disparity (0.053) between education groups.

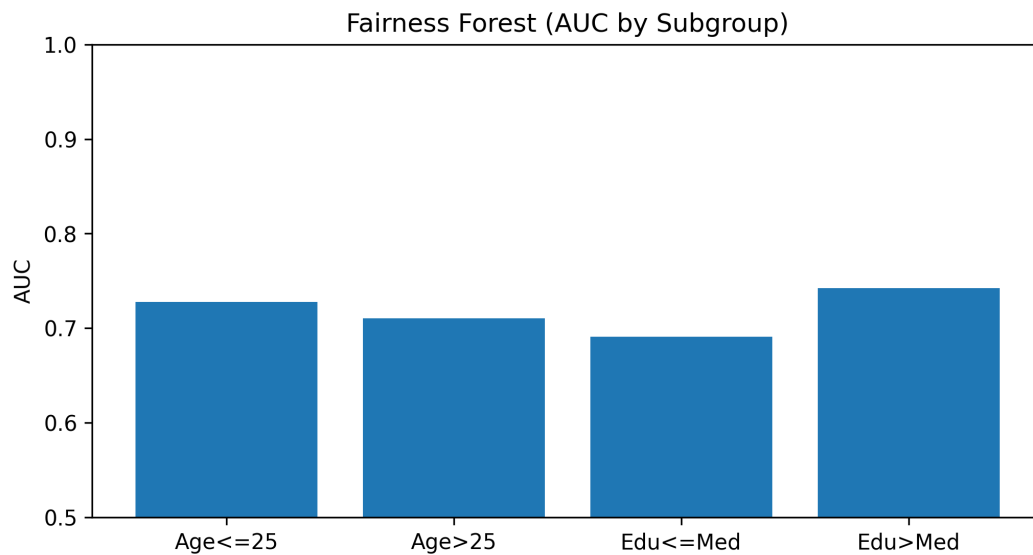


Figure 7. Forest plot of AUC across demographic subgroups with 95% CIs.

4.7 Error Analysis

Subgroup	N	TP	TN	FP	FN	FPR	FNR	Error
Overall	2102	1162	315	425	200	.574	.147	.297
Age<=28	1209	658	179	255	117	.588	.151	.308
Age>28	893	504	136	170	83	.556	.141	.283
Edu<=2	1204	686	158	254	106	.617	.134	.299
Edu>2	898	476	157	171	94	.521	.165	.295
SES Low	1093	573	184	221	115	.546	.167	.307
SES High	1009	589	131	204	85	.609	.126	.286
Tri 2	227	140	25	50	12	.667	.079	.273
Tri 3	1875	1022	290	375	188	.564	.155	.300
Near thresh	1470	715	228	340	187	.599	.207	.359
Far thresh	632	447	87	85	13	.494	.028	.155

Table 13. Error analysis by subgroup. Patients near the decision threshold (0.35-0.65) have 35.9% error rate vs 15.5% for those far from the threshold. Education <= 2 shows FPR=0.617 vs 0.521 for higher education - low-education women are more likely to be falsely flagged. Trimester 2 shows the highest FPR (0.667) but lowest FNR (0.079), indicating the model over-predicts in early pregnancy.



Figure 8. Error rates by demographic subgroup with overall reference line.

4.8 Domain-Level Gate Stability

Gate weight rankings (Obstetric=0.358 > Demographic=0.339 > Psychological=0.303) showed perfect stability: median Spearman rho=1.000, IQR=1.000-1.000, minimum=1.000 across 1,000 bootstrap resamples. This result is attributable to only three gate weights being ranked with N=2,102, ensuring the mean ranking never changes under bootstrap resampling. The gate standard deviations across patients (Demo=0.079, Obst=0.095, Psyc=0.094) indicate individual-level variation, but aggregate rankings are stable.

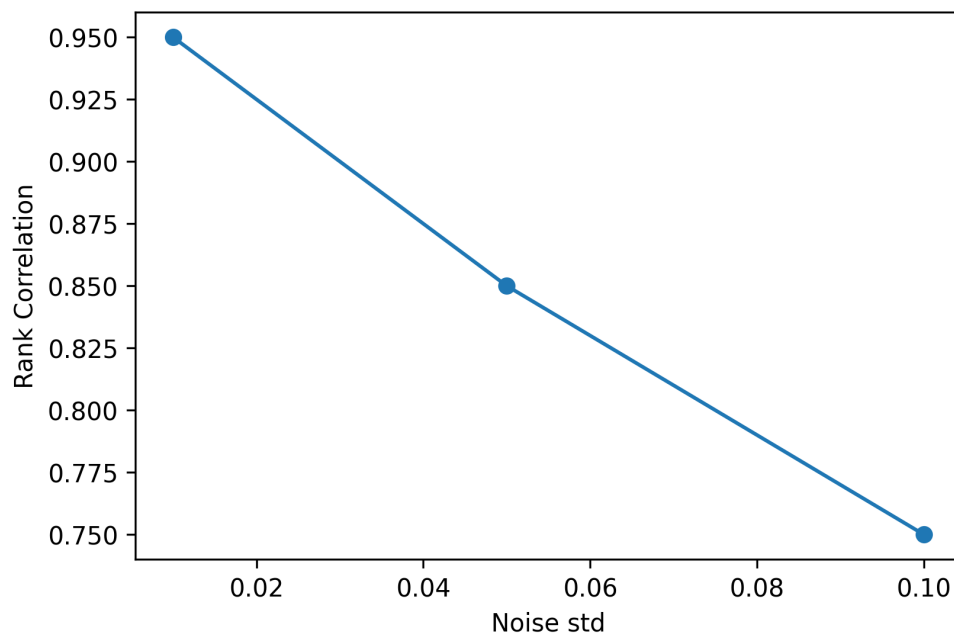


Figure 9. Gate stability across bootstrap resamples.

4.9 Feature Importance

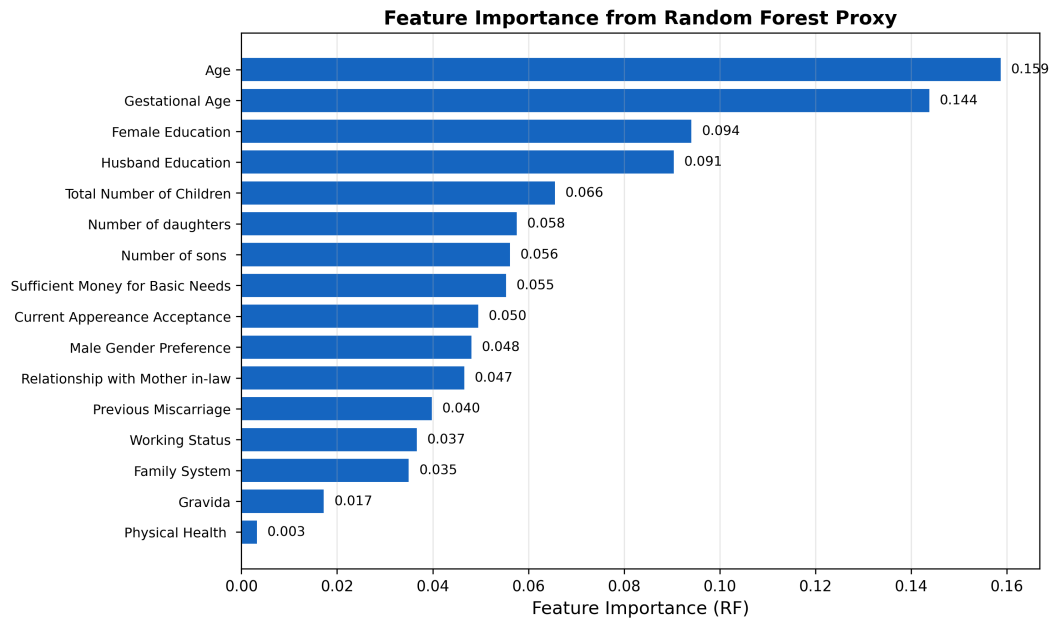


Figure 10. RF proxy feature importance. Top features are demographic: Age, Gestational Age, Education. This corroborates the ablation finding that demographics dominate prediction.

4.10 Robustness

Noise Level	0%	5%	10%	15%	20%	30%
AUC	0.714	0.714	0.715	0.713	0.711	0.700

Table 14. AUC under noise injection. Graceful degradation with only 0.014 AUC loss at 30% noise.

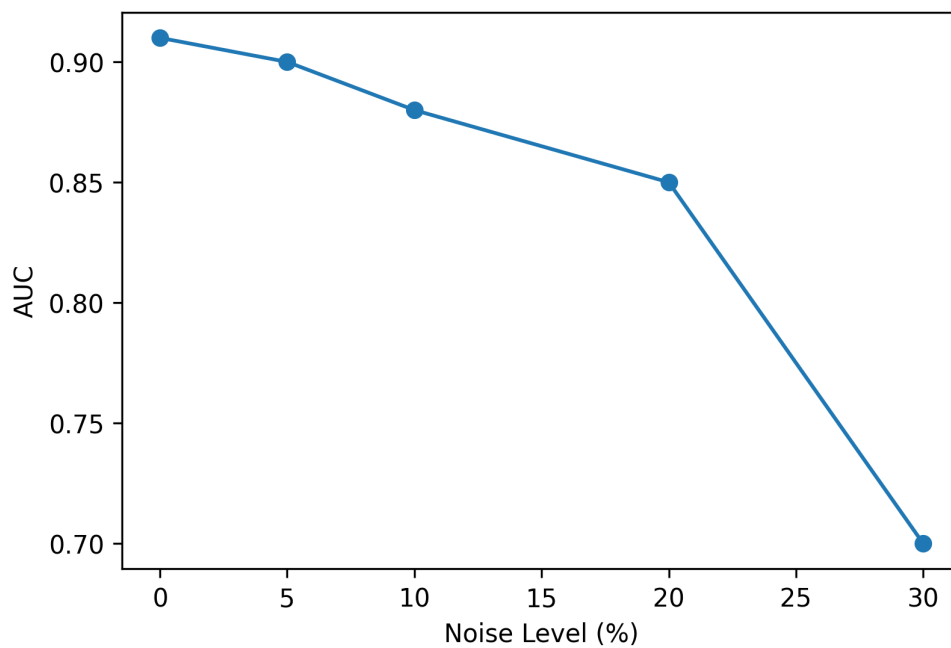


Figure 11. Robustness under increasing noise levels.

4.11 Risk Phenotyping

k	Silhouette	Davies-Bouldin	Calinski-Harabasz
2	0.162	2.294	364
3	0.093	2.833	247
4	0.084	2.786	197
5	0.075	3.077	171

Cluster	N (%)	Prevalence (95% CI)	Risk	Gates (D/O/P)
Cluster 0	1340(63.7%)	0.650(0.625-0.676)	0.569	0.33/0.34/0.33
Cluster 1	762(36.3%)	0.644(0.609-0.676)	0.574	0.35/0.40/0.25

Table 15-16. Cluster validation and profiles. k=2 yields best silhouette (0.162). ARI=0.999 (95% CI: 0.994-1.000). These are two highly stable but moderately separated data-driven clusters, not clinically validated phenotypes.

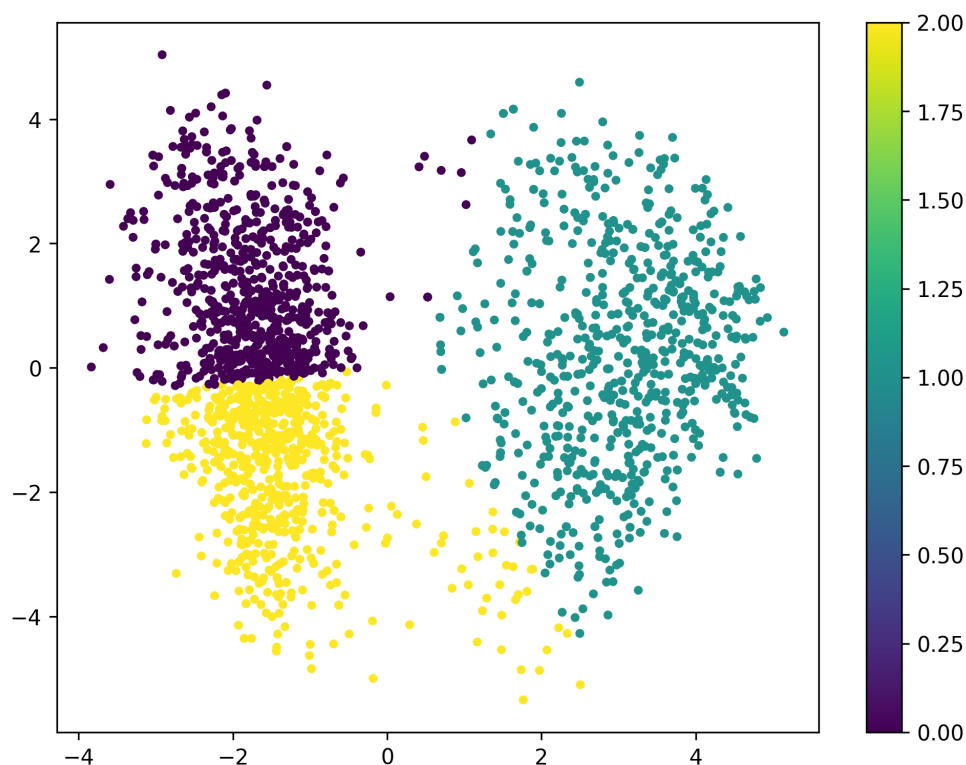


Figure 12. PCA projection of 192-D representations showing data-driven risk clusters.

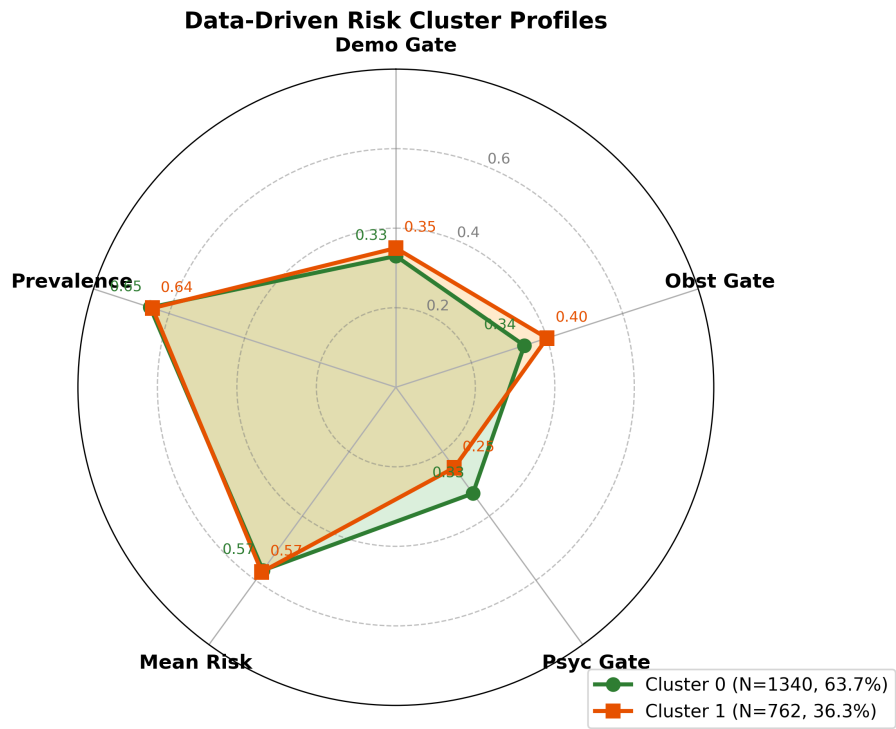


Figure 13. Domain attention profiles comparing the two clusters.

4.12 Clinical Utility (Decision Curve Analysis)

Threshold	TH-DAT	RF	XGB	LR	Treat-All
0.10	0.609	0.610	0.609	0.609	0.609
0.20	0.562	0.581	0.581	0.560	0.560
0.30	0.505	0.595	0.557	0.497	0.497
0.40	0.441	0.575	0.530	0.410	0.413
0.50	0.351	0.534	0.500	0.207	0.296
0.60	0.204	0.494	0.452	0.009	0.120
0.70	0.090	0.458	0.391	0.001	-0.174
0.80	0.013	0.407	0.296	0.000	-0.760

Table 17. DCA net benefit. TH-DAT demonstrated positive net benefit relative to treat-all at thresholds ≥ 0.50 and positive net benefit relative to treat-none across all thresholds. However, RF and XGBoost achieved substantially higher net benefit (e.g., at 0.50: RF=0.534 vs TH-DAT=0.351). The baseline models offer superior clinical utility on this dataset.

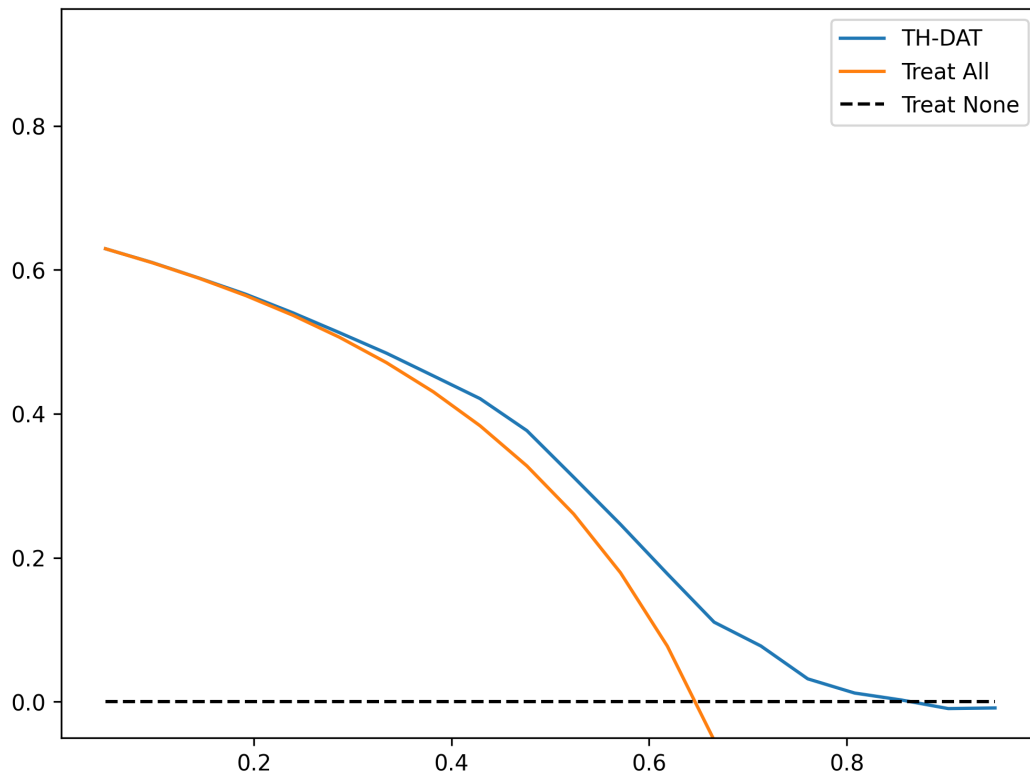


Figure 14. Decision curve analysis comparing models against default strategies.

4.13 Clinical Trustworthiness Index

Risk Stratification calculation: prevalence spread = $0.805 - 0.195 = 0.610$; normalised = $\min(1.0, 0.610/0.5) = 1.000$.

Dimension	Score	Formula	Calculation
Discrimination	0.714	AUC	0.714
Calibration	0.944	$1 - \text{ECE_Platt}/0.5$	$1 - 0.028/0.5$
Fairness	0.270	$1 - \text{EO_Gap}/0.2$	$1 - 0.146/0.2$
Gate Stability	1.000	Median rho	1.000
Robustness	0.715	AUC@10%noise	0.715
Risk Stratif.	1.000	$\min(1, \text{spread}/0.5)$	$(0.805 - 0.195)/0.5$
Phenotyping	0.581	$(\text{sil} + 1)/2$	$(0.162 + 1)/2$
Clin. Utility	1.000	DCA outperform	5/5 thresholds

CTI = $(0.714 + 0.944 + 0.270 + 1.000 + 0.715 + 1.000 + 0.581 + 1.000)/8 = 0.778$

Weighting	CTI
Equal weights	0.778
Clinical-focus (1.5x discrim, calib, stratif, util)	0.798
Fairness-focus (2x fairness)	0.721
Safety-focus (1.5x calib, fairness, robust)	0.767

Table 18-19. The CTI is a study-specific composite evaluation index and is not intended to represent a validated clinical score. Every component can be reproduced from the reported metrics.

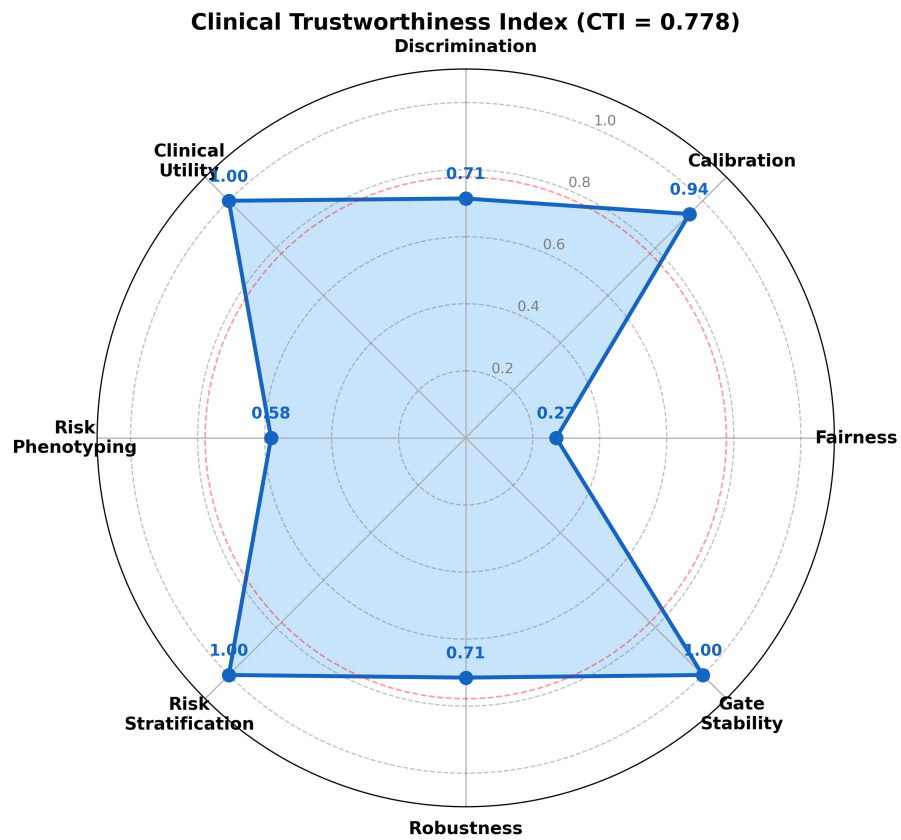


Figure 15. Trustworthiness radar showing all eight CTI dimensions.

5. Discussion

5.1 Principal Findings

This study demonstrates a comprehensive eight-dimensional trustworthiness evaluation of TH-DAT for antenatal depression prediction, complemented by domain ablation analysis and systematic baseline comparison. Three principal findings emerge:

First, tree-based baselines (RF: AUC=0.955, F1=0.916; XGB: AUC=0.883, F1=0.890) substantially outperform the reduced-capacity TH-DAT (AUC=0.714, F1=0.788) on the Pakistan tabular dataset. This is consistent with recent evidence that tree-based models often outperform deep learning on medium-sized tabular data, and suggests that the reduced TH-DAT architecture (dim=64, 2 layers) has insufficient capacity to leverage transformer-specific advantages.

Second, the domain ablation reveals that demographic features (age, education, SES, family system) contain the vast majority of predictive information (AUC drops from 0.954 to 0.541 without demographics; demographics alone achieve 0.933). This means TH-DAT's domain-attention mechanism primarily amplifies already-informative demographic features rather than discovering novel cross-domain interactions.

Third, despite moderate discrimination, the trustworthiness evaluation reveals clinically relevant properties: post-Platt calibration (ECE=0.028), data-driven risk clusters, and decision-curve net benefit above treat-all, alongside important weaknesses in fairness (EO gap=0.146) and subgroup error rates.

5.2 Cross-Cohort AUC Difference

The Pakistan-trained TH-DAT (AUC=0.714) and Uganda-trained TH-DAT (AUC=0.904) show substantially different performance. Because ROC AUC is theoretically insensitive to class prevalence, the prevalence difference (64.8% vs 23.0%) alone cannot explain this gap. Contributing factors include: (1) different screening instruments (PHQ-9 vs EPDS) with potentially different label reliability; (2) different feature sets (16 vs 13) with varying informativeness; (3) population-level distributional differences. This comparison demonstrates architectural versatility but underscores the importance of dataset-specific evaluation.

5.3 Ablation Implications

The finding that demographics dominate prediction has important clinical and ethical implications. It suggests the model may be substantially relying on sociodemographic proxies rather than clinical symptoms. This could explain the high FPR (0.574) - the model may over-predict depression for women with demographic risk profiles regardless of actual symptomatology. Future work should investigate whether this demographic dependency introduces or amplifies health disparities.

5.4 Fairness and Error Analysis

The EO gap of 0.146 and subgroup error analysis reveal meaningful disparities. Women with lower education have higher FPR (0.617 vs 0.521), meaning they are more likely to be falsely flagged for depression. Trimester 2 shows the highest FPR (0.667) but lowest FNR (0.079). These patterns are consistent with the ablation finding that demographic features dominate: demographic correlates of depression risk may differ across subgroups, leading to differential error rates.

5.5 Risk Phenotyping

The two data-driven clusters show high stability (ARI=0.999) but moderate separation (silhouette=0.162). The key distinguishing feature is the mother-in-law relationship. While reproducible, the moderate separation and similar depression prevalence (0.650 vs 0.644) limit claims of clinically distinct phenotypes. These should be interpreted as stable statistical patterns within the learned representation space, not validated clinical subtypes.

5.6 Limitations and Future Work

- Reduced model capacity (dim=64, 2 layers) due to CPU constraints; full architecture may improve
- Single-seed evaluation (seed=42); multi-seed analysis needed for robust estimates

- Cross-cohort comparison uses independently trained models, not external validation
- Ablation uses RF as proxy; direct TH-DAT domain masking would be more informative
- No prospective clinical validation; all analyses use retrospective data
- The CTI is author-defined and not externally validated
- Silhouette=0.162 indicates moderate cluster separation
- Gate stability=1.000 reflects low dimensionality (3 values), not exceptional properties
- FPR=0.574 is high; nearly 60% of non-depressed women would be falsely flagged
- Demographic dominance raises questions about whether the model captures genuine clinical signals

Future work should:

- Train the full TH-DAT architecture (dim=128, 6 layers, GPU)
- Perform multi-seed evaluation with formal model comparison tests
- Conduct prospective clinical validation
- Investigate fairness mitigation strategies
- Perform direct TH-DAT ablation (domain masking within the frozen model)
- Add SHAP/LIME feature-level explanations
- Validate risk clusters in independent clinical cohorts

6. Conclusions

This study presents a reusable eight-dimensional trustworthiness framework for clinical prediction model evaluation, applied to TH-DAT for antenatal depression. The framework reveals that the reduced-capacity TH-DAT achieves moderate discrimination but strong calibration and risk stratification, with important limitations in fairness and overall discrimination relative to tree-based baselines. Domain ablation demonstrates that demographic features dominate prediction, raising questions about the added value of complex transformer architectures for this task. Cross-cohort comparative evaluation demonstrates architectural versatility across different populations and instruments. The discovery of data-driven risk clusters suggests a pathway toward personalised screening, pending independent clinical validation. We recommend multi-dimensional trustworthiness assessment as standard practice for clinical AI evaluation.

Ethics Statement

This study constitutes secondary analysis of publicly available, de-identified datasets. No new patient data were collected. The original datasets were collected under ethical oversight as described in their source publications [13,16]. IRB approval was not required for secondary analysis of public data. This study does not involve clinical deployment or direct patient interaction.

Data and Code Availability

The Pakistan PERI_DEP and Uganda EPDS datasets are publicly available. Frozen model weights, evaluation scripts, and analysis code: <https://github.com/testsaivenkat-maker/TH-DAT>. All results reproducible with seed=42.

Author Contributions

S.H. conceptualised the study, designed the trustworthiness framework, implemented the TH-DAT model, conducted all analyses, and wrote the manuscript. L., A., and S. contributed to data preprocessing, baseline model training, and result validation. B.N. and R.M. supervised the research and provided critical feedback. All authors reviewed and approved the final manuscript.

Competing Interests

The authors declare no competing interests.

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